

HELIOLITE · DESIGN DECISION RECORD

HelioLite recommendations accepted

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-26	Amish Chadha	Budget top-up to \$455 decided by Amish (P1)
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); record the TRL 3 review items now decided, what changed in the repo and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted for items N1 to N6; items O1 to O3 and P1 remain proposed

■ Context

The TRL 3 review (docs/REVIEW.md , session 2026-09-25: TRL 3, and HLT-DDR-001, item O4) listed six new items as "Proposed, awaiting Amish", each with a recommendation. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every item with a recommendation is therefore decided in favor of that recommendation; where a recommendation named one of several options, that option is the decision. Items without a recommendation stay open. TRL 4 work stays on hold by Amish's instruction, and HelioLite stays at trl: 3, trl_target: 3.

The ten TRL 2 items (D1 to D10) were already decided in HLT-DDR-001 and are unchanged.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
N1	Stowed wind survival (R9)	Option B: stow stops on the yoke that carry the stowed hinge moment in both directions, checked against option C (published stowed coefficients), rather than a larger NMRV040-class gearbox (A)	Because a fixed stop can resist only one direction on an axis that turns into the stow, the stops are a polyurethane stop pad below a steel lug on the torque tube and a sprung latch pawl above it, on a steel bracket on the -X arm, with a 12 V pull solenoid to release the pawl. Designed for twice the assumed stowed moment (48.6 N·m, a coefficient up to 0.30), which is the check against C. Added to <code>cad/src/model.py</code> (part 17), BOM line 17 (\$20), HLT-PRC-001, HLT-CAL-001 [E7, E8] and drawing HLT-DWG-001 Rev P2. R9 moves from not met (24.3 N·m on a gearbox with a 22 N·m listed maximum) to met on paper
N2	Drive preload (R4)	A spiral preload spring of about 3 N·m on each axis, about \$6 for both	Model part 16 (spring cans on the -X trunnion and inside the mast top), BOM line 16 (\$6), pointing budget now counts on it. The elevation spring biases the mirror toward stow. R4 stays at risk (0.31° typical, 0.50° at the 95th percentile of calibration)
N3	Mass limit (R13)	Option (a): relax R13 from 10 kg to 13 kg, rather than lighter drives (b) or aluminum arms (c)	HLT-REQ-001 R13 reads 13 kg. With N1 and N2 the mast-top mass is 12.95 kg, so R13 moves from not met (12.5 kg against 10 kg) to met with 0.05 kg margin on assumed masses
N4	R7 wording	"30 min or less of hands-on time, spread over one clear day"	HLT-REQ-001 R7 reworded; R7 moves from at risk to met (19 min hands-on over about 4 h)
N5	R10 wording	During a stow the beam may move only downward, toward the ground	HLT-REQ-001 R10 reworded; R10 moves from at risk to met (stow in 23 s, beam never above -4°)
N6	Calibration points	Four points over about 4 h as the default, within D7's "three or more"	HLT-PRC-001 and HLT-CAL-001 use four points over about 4 h as the default (95th percentile residual 0.23° on the normal)

Consequences for the budget: the recommendations for N1 and N2 carried no budget figure. Lines 16 and 17 take the BOM from \$425 to \$451 against the \$430 in `project.yaml`, so R15 moves from met to **not met**. `budget_usd` stayed at \$430 until P1 below was decided.

Budget top-up to \$455: decided by Amish, 2026-09-26 (P1, option a). `budget_usd` \$430 to \$455; HLT-REQ-001 v0.5 and HLT-CAL-001 v0.3: R15 moves from not met to met (\$451, \$4 margin).

Table 2. Items still open (Proposed, awaiting Amish).

#	Item	Why it stays open
O1	First site and user for the co-design checklist (house, greenhouse or school)	No recommendation was made; co-design partners stay open by Amish's instruction
O2	Local rules on glare and structures at the first site	Depends on O1
O3	Siting survey of real yards	No recommendation was made
P1	Budget for the \$451 BOM (R15)	New. Options: (a) raise <code>budget_usd</code> to \$455; (b) keep \$430 and cut cost elsewhere, for example print the turntable and source a cheaper anemometer; (c) treat the anemometer as optional where a reliable forecast is available. Recommendation: (a), because both additions are safety or accuracy items, as with D4. Decided by Amish, 2026-09-26: budget top-up to \$455 (option a)

■ Consequences

- HLT-PRB-001 is unchanged. HLT-PRC-001 and HLT-REQ-001 move to v0.4, HLT-CAL-001 to v0.2, drawing HLT-DWG-001 to Rev P2; the model, STEP, STL and media were regenerated.
- Requirement status: 11 met, 1 not met (R15), 2 at risk (R3, R4), 1 not verifiable at TRL 3 (R12). After the 2026-09-26 budget top-up (P1): 12 met, 0 not met, 2 at risk, 1 not verifiable.
- No cross-repo action arises from these items.
- The latch, springs and solenoid are paper design only. Building and testing them (holding moment, pad compliance, latch engagement on a power-loss stow) is TRL 4 work and stays on hold by Amish's instruction.